Jiajin (David) Wu

AI/LLM Software & Application Engineer

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Summary

AI/LLM software and application engineer with 3+ years building production Python/TypeScript systems for enterprise workflows. At Aptiv, delivered 3 workflow agents and a governed multi-agent platform spanning LangGraph, hybrid retrieval-augmented generation (RAG), human-in-the-loop controls, FastAPI, and a React/Next.js operator console. Technical lead who directed 430+ issues across 9 L1–L3 ADAS programs and earned a 121% top-tier performance rating for two consecutive years.

Technical Skills

AI/LLM Engineering: LangGraph, LangChain, deepagents, RAG, tool calling, LLM evaluation, prompt engineering, OpenAI, Anthropic Claude, Google Gemini, human-in-the-loop
Languages & Frameworks: Python, TypeScript, React, Next.js, FastAPI, Pydantic, SQLAlchemy, SQL, MATLAB
Data & Retrieval: OpenSearch, BM25, embeddings, kNN/vector search, hybrid search, SQLite, MinIO/S3, semantic memory
Platform & Integration: REST APIs, asyncio/aiohttp, JWT/RBAC/OAuth, Jenkins CI/CD, Docker, Microsoft Graph, Jira, Polarion, Confluence, pytest, Ruff
Systems Engineering: ADAS L1–L3, CAN/CANoe, MBSE, SysML, MagicDraw/MagicGrid, requirements, verification, root-cause analysis

Work Experience

Vehicle System Triage Engineer — Technical Lead, VSDA Triage Team — **Feb 2023 — Present**
Aptiv Corporation — *Troy, MI*

- Built and deployed 3 Python AI/LLM agents: Release Email Analyzer extracts software versions/change sets and triggers stream-definition builds; VSDA Email Agent converts VSI/VSDA reports into release reports; Jira Ticket Extractor populates the triage database
- Engineered a Jira REST API ticket-automation tool that generated 1,000+ standardized stories team-wide; automated work-history reporting with Jira APIs and Jenkins for near-real-time management visibility
- Led cross-functional issue triage across 9 L1–L3 ADAS programs, personally triaging 267 and directing 430+ system-, vehicle-, and bench-level tickets across Algorithm, Integration, Testing, and Data-Mining teams
- Built reusable Python/MATLAB batch pipelines for large AEB/PEB datasets, calculating camera/module latency and comparing AHBC performance before/after coating changes to deliver customer-ready analysis
- Automated Polarion test-plan generation, test-case import, and result updates; authored 38 integration and radar-signal test cases with end-to-end requirements traceability
- Earned a 121% top-tier performance rating with "Exceeds Expectations" across all goals for two consecutive years; represented the team in customer meetings and company-wide AI working sessions

Graduate Student Instructor — Systems Engineering (ISD 521) — **Aug — Dec 2022**
University of Michigan, College of Engineering — *Ann Arbor, MI*

- Co-guided 51 students across 13 teams through a four-deliverable MBSE project worth 40% of the course grade, translating stakeholder needs into SysML context, behavior, traceability, architecture, and validation models in MagicDraw/MagicGrid
- Authored a five-slide technology-readiness recitation and an 83-formula Excel decision-tree model comparing 3 investment strategies through expected-payoff and value-of-information analysis

Systems Engineering Intern — **Jun — Sept 2022**
Varian Medical System, a Siemens Healthiness Company — *Palo Alto, CA*

- Designed a MATLAB GUI application that generated parameterized XML test plans, streamlining execution of radiotherapy-system verification workflows
- Built MATLAB pipelines to acquire, process, and visualize beam data for control-system diagnosis and Beam Generation Monitor subsystem verification
- Collaborated with mechanical, electrical, and software engineers to test a next-generation radiotherapy treatment-system prototype and document results

Additional Work Experience

Mechanical Engineering Co-op **May — Aug 2021**
Contemporary Intelligent Manufacturing Co. Ltd. *Ningde, China*

- Led development and production-line deployment of a portable, non-contact electrode-thickness measurement system meeting a $10,000 maximum-cost requirement and a 0.2-micrometer resolution target

Systems Engineering Intern **Mar — Jun 2020**
Shanghai MAXIEYE Technology Co. Ltd. *Shanghai, China*

- Developed L2 collision-warning requirements and verification plans, then wrote MATLAB tooling to clean and visualize CAN signals for system validation
- Analyzed NHTSA human-factors guidance for L2+/L3 ADAS development and translated findings into system-level engineering inputs

Selected AI/LLM Projects

VSDA Deep Agent — Full-Stack Enterprise AI Platform **Python, LangGraph, FastAPI, OpenSearch**

- Architected 8 deployable LangGraph entry points—1 supervisor and 7 domain subagents—exposing 60 tools through 46 authenticated FastAPI routes for Jira, Teams, email, OpenSearch, Polarion, Confluence, and Jenkins
- Built hybrid BM25/kNN RAG, durable ingestion from 68 source manifests, semantic memory, role-based access, human approval, and quota enforcement; backed the platform with 2,132 test functions across 266 files
- Built a React 19/Next.js 16 operator console with concurrent token/state streaming, 3 role tiers, multimodal uploads, model governance, and approve/edit/reject controls for agent tool calls

Resume Tailor Harness — Fact-Locked AI Job Pipeline **Python, FastAPI, React, TypeScript**

- Built a full-stack application and Typer CLI that ingests resumes and GitHub repositories, scores postings across 18 connector kinds, and tailors resumes and cover letters through multi-agent review
- Unified OpenAI, Anthropic, Google, and DeepSeek behind one model abstraction; implemented bounded async fan-out, multi-tenant workspaces, OAuth/PBKDF2 authentication, per-user SQLite isolation, and SSRF-hardened fetching

Enterprise Engineering Automation Suite **Python, REST APIs, Jenkins, Jinja2**

- Built async enterprise clients spanning 13 Jira endpoints, 62 Polarion methods, 19 typed ALM domain models, and Confluence/Microsoft Graph integrations with concurrent pagination and pooled HTTP sessions
- Delivered 5 scheduled Jira report entry points, checkpointed Jenkins stages, batch Polarion edits/PDF exports, and YAML/Jinja2 pipelines for Confluence publishing and Outlook stakeholder communications

Education

Master of Engineering, Systems Engineering & Design — University of Michigan–Ann Arbor Dec 2022, GPA: 3.9
Bachelor of Science, Mechanical Engineering — Shanghai Jiao Tong University Aug 2021, GPA: 3.6

Publications

M. Liu, F. Zhao, **J. Wu**, W. Fan, J. Zhang, D. Hung, *Hybridizing Unsupervised Clustering Methods for In-cylinder Vortex Motion Analysis Under Different Swirl Ratio Conditions*, SAE WCX, 2021
J. Wu, J. Wang, Y. Yang, Z. Zeng, D. Lu, and L. Lian, *Hybrid Aerial-Aquatic Vehicle for Large Scale High Spatial Resolution Marine Observation*, OCEANS, 2019